ME21 Project 1 Deliverable

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Tracker Software Analysis:

Motor RPM Calculation:

RPM Calculation:

Row 189: $t = 6.267s$, $x = -3.274E-2\,m$, $y = -0.123m$

Row 414: $t = 13.767s$, $x = -3.278E-2\,m$, $y = -0.125m$

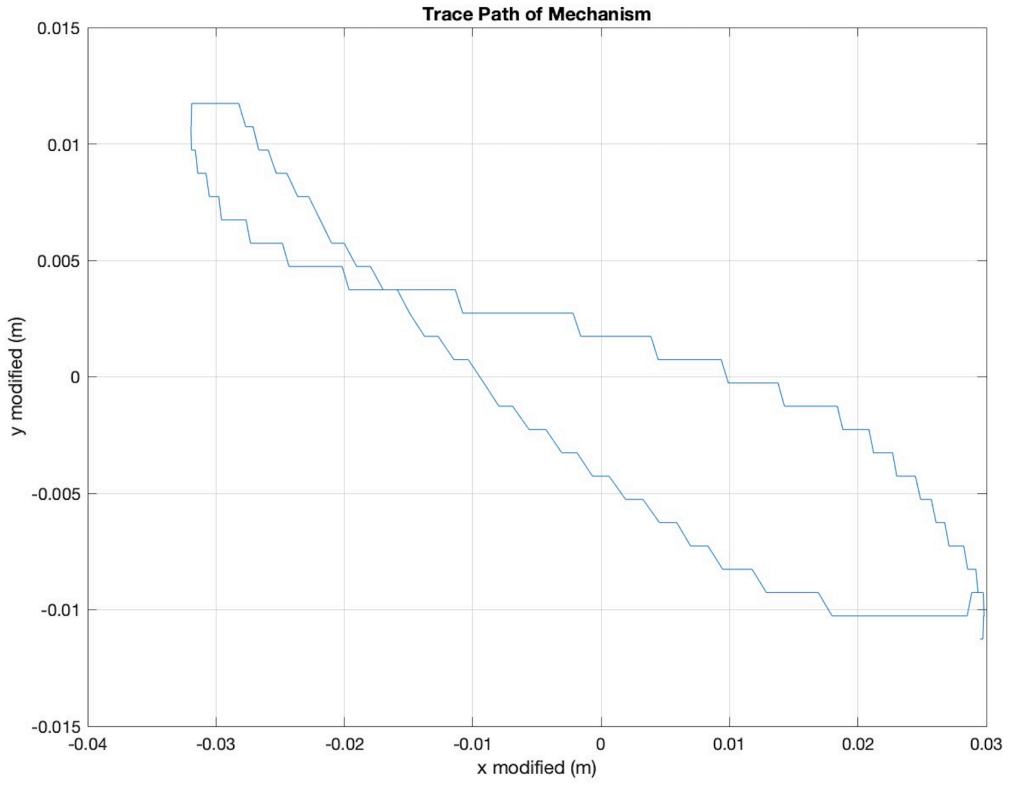
$\Delta t = 7.5s$ per cycle

$$\frac{\text{fps slow}}{\text{fps normal}} = \frac{240}{30} = 8$$

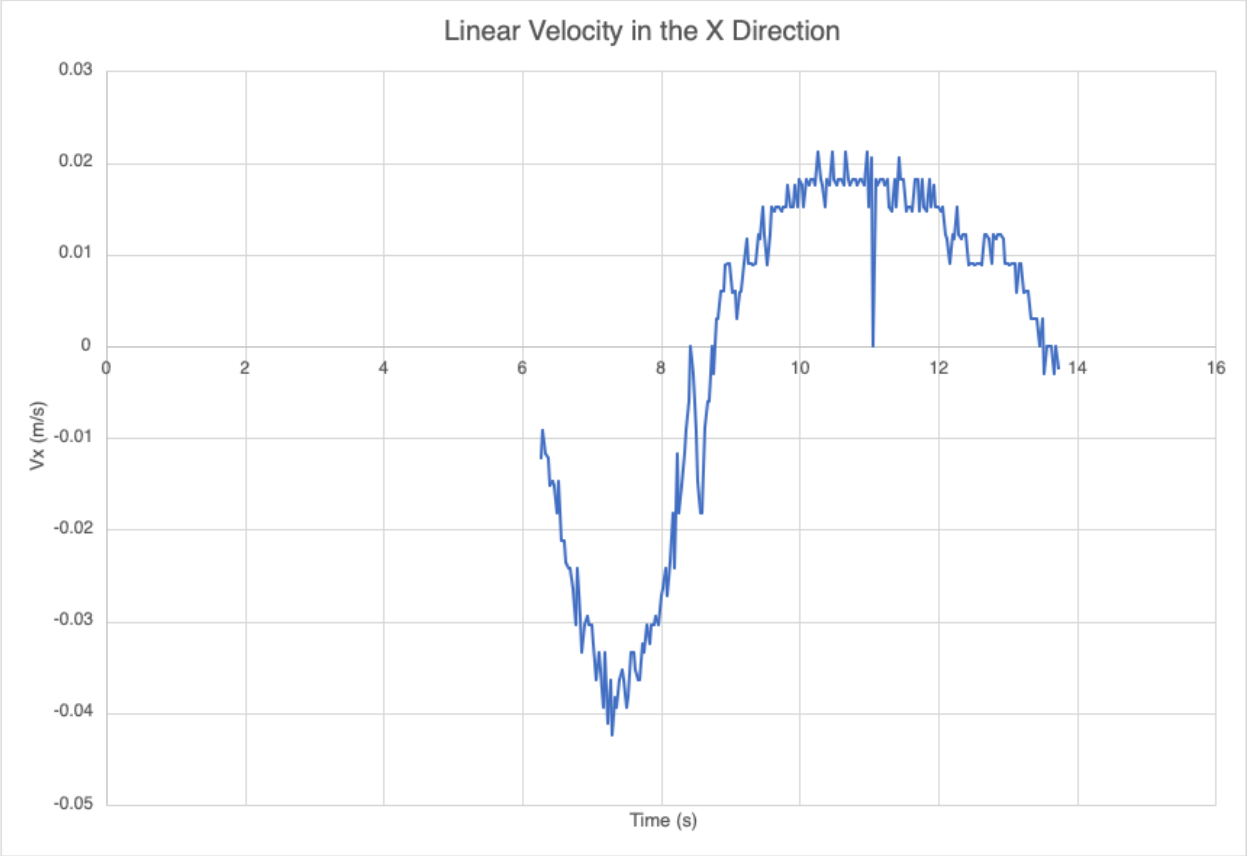
$$\frac{7.5s}{8} \cdot 60 = 56.25 \text{ cycles/min}$$

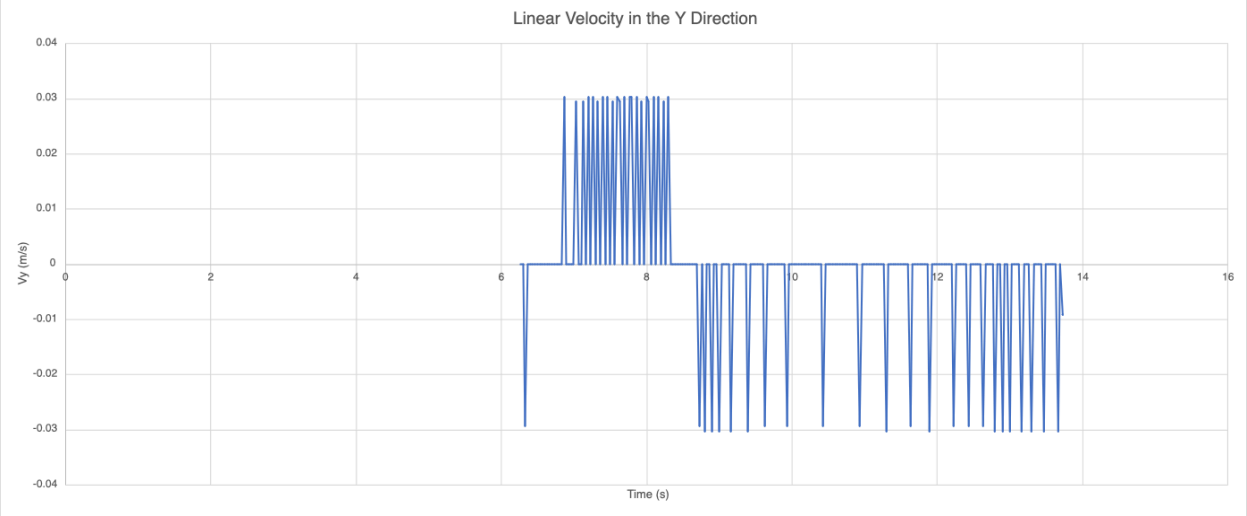
So motor rotates at approximately 56.25 rpm

Path Trace (for a full cycle):



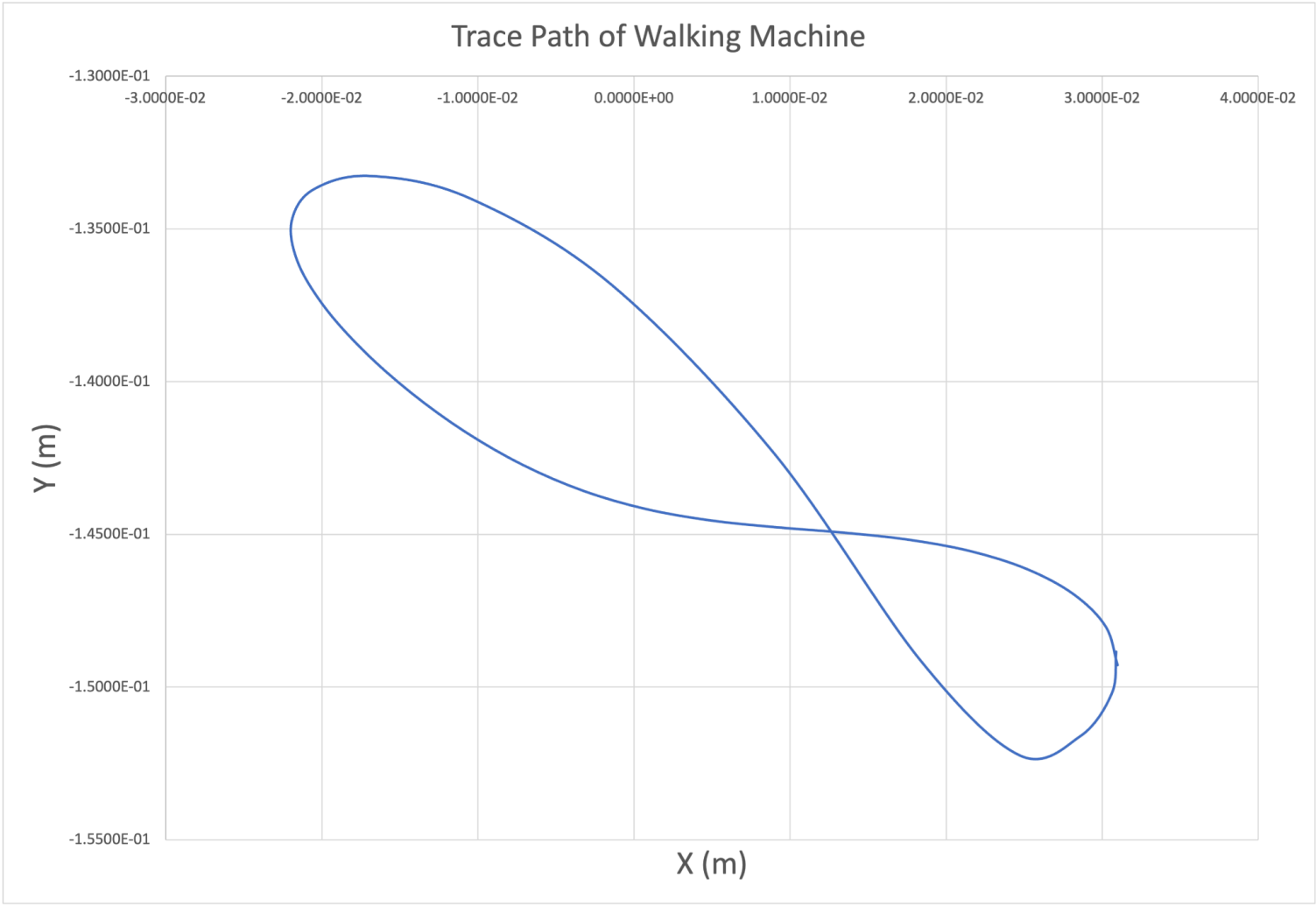
Velocity in the X and Y Directions



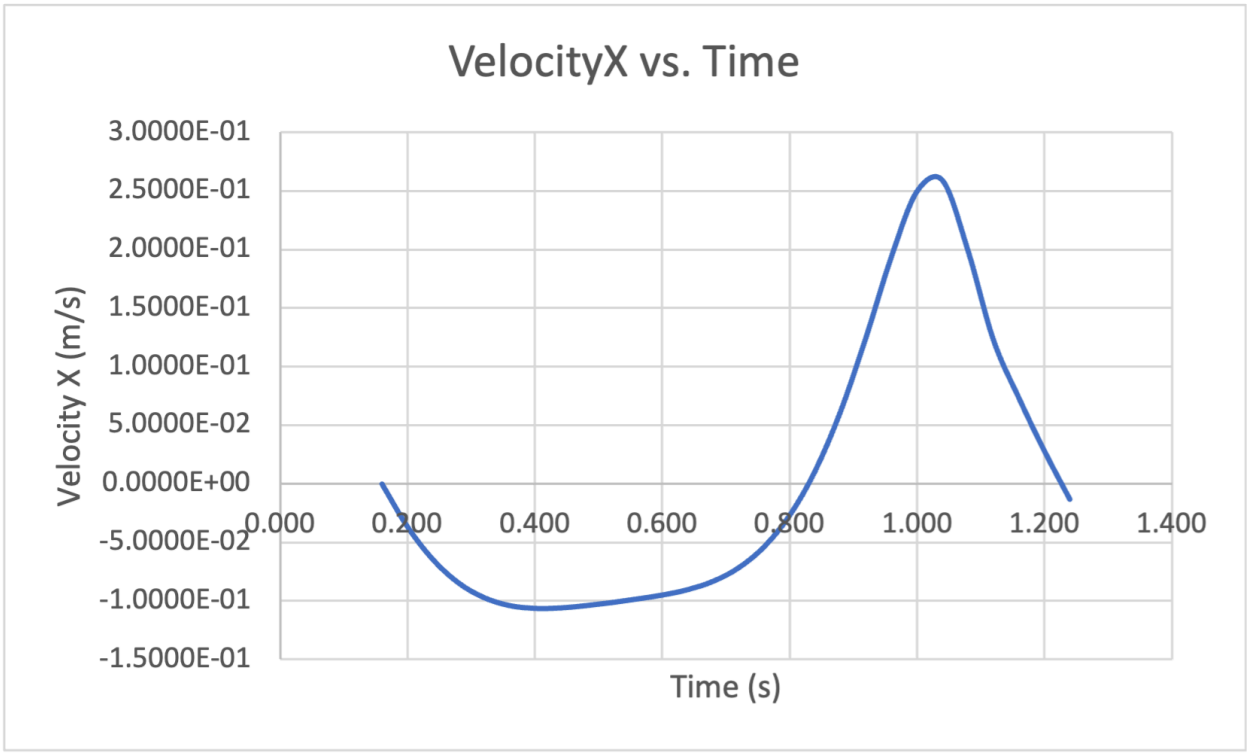


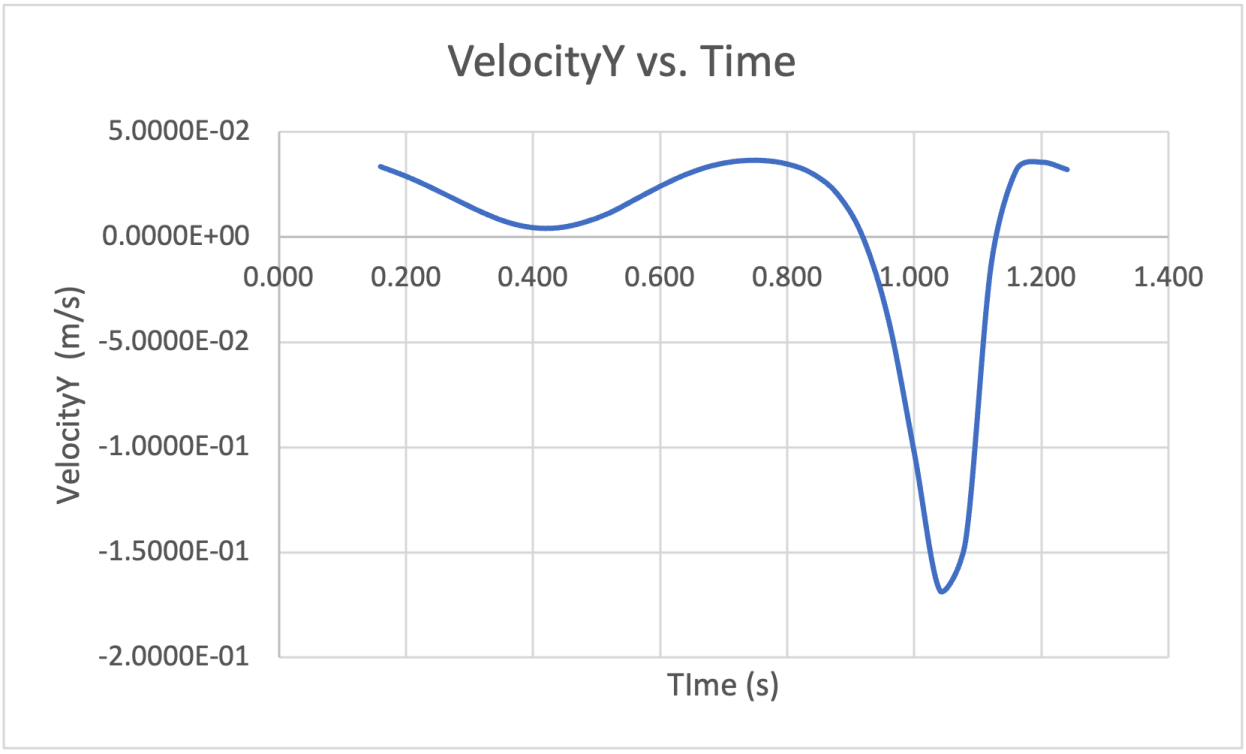
SolidWorks Analysis

Path Trace (for a full cycle):



Velocity in the X and Y Directions





Differences Between Tracker and SolidWorks Plots

The major differences between the tracker and SolidWorks plots are in the velocity graphs. The path trace is fairly similar, although the SolidWorks path trace is much smoother. We think that this may be due to the fact that we widened the tracking window on tracker so that it would automatically follow the point on our video properly. This made the tracker plot less continuous than the SolidWorks one. However, the velocity plots look quite different. If one sketches the overall shape, both the tracker and SolidWorks velocity plots follow similar curves, however, the SolidWorks data provides a much smoother curve, especially in comparison to the y-velocity. The x-velocity graph from the tracker data matches fairly well, however the initial curve is much more shallow and the second peak is much steeper in the SolidWorks plot. The y-velocity is very choppy based on the tracker data. In the tabulated data from tracker, it can be seen that, especially for the y-velocity, there are some discontinuities where the graphs go to zero, which could be errors in the tracker software. This caused deviations in our graphs from the curves given by SolidWorks, especially for the vertical displacement which frequently returns to the x-axis and oscillates very irregularly. However, both the SolidWorks and tracker plots of the velocities show that x and y have a fairly inverse relationship. Overall, the plots from tracker follow a similar shape to the plots from SolidWorks, but the data from SolidWorks seems to provide much more clear and intuitively accurate results and plots.